INTERNATIONAL MONETARY FUND

CHAD

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REQUEST FOR A THREE- YEAR ARRANGEMENT UNDER THE EXTENDED CREDIT FACILITY—DEBT SUSTAINABILITY ANALYSIS

Approved By
Vitaliy Kramarenko, Gavin Gray (IMF), Abebe Adugna, Marcello Estevão (IDA)

Prepared by the Staff of the International Monetary Fund (IMF) and the International Development Association (IDA).

Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*In debt distress*[1]
Overall risk of debt distress	*In debt distress*
Granularity in the risk rating	*Unsustainable*
Application of judgement	*No*

Chad's debt ratios have worsened since the onset of the COVID-19 pandemic and downside risks have increased given the uncertainties around the intensity and duration of the pandemic and oil price volatility. On the external debt side, liquidity pressures have increased given a downward revision in exports in 2020 which implied lower oil revenues. As a result, in the pre-restructuring DSA, the external debt service to revenue ratio breaches its threshold under the baseline over a prolonged period. Identified domestic and external funding sources are insufficient to meet Chad's financing needs over the medium to long term, even with an ambitious fiscal consolidation. Altogether, Chad's public debt is assessed to be unsustainable. Considering these financing and debt challenges, on December 14, 2020, the authorities announced their intention to seek debt restructuring under the G20 Common Framework. On the domestic debt side, the authorities are in the process of rolling over the current debt stock to longer maturities. In view of the ongoing debt restructuring negotiations, Chad is assessed as being in external and overall debt distress. The ongoing debt restructuring is expected to restore debt sustainability and bring the risk rating to 'moderate' within the program period.

[1] With a score of 2.47, Chad's composite indicator, which is based on the October 2020 WEO and the 2019 CPIA, signals a weak debt-carrying capacity. Thus, the relevant external debt burden high-risk thresholds are: (i) 30 percent for the present value (PV) of external debt-to-GDP ratio; (ii) 140 percent for the PV of debt-to-exports ratio; (iii) 10 percent for the debt service-to-exports ratio; and (v) 14 percent for the debt service-to-revenue ratio.

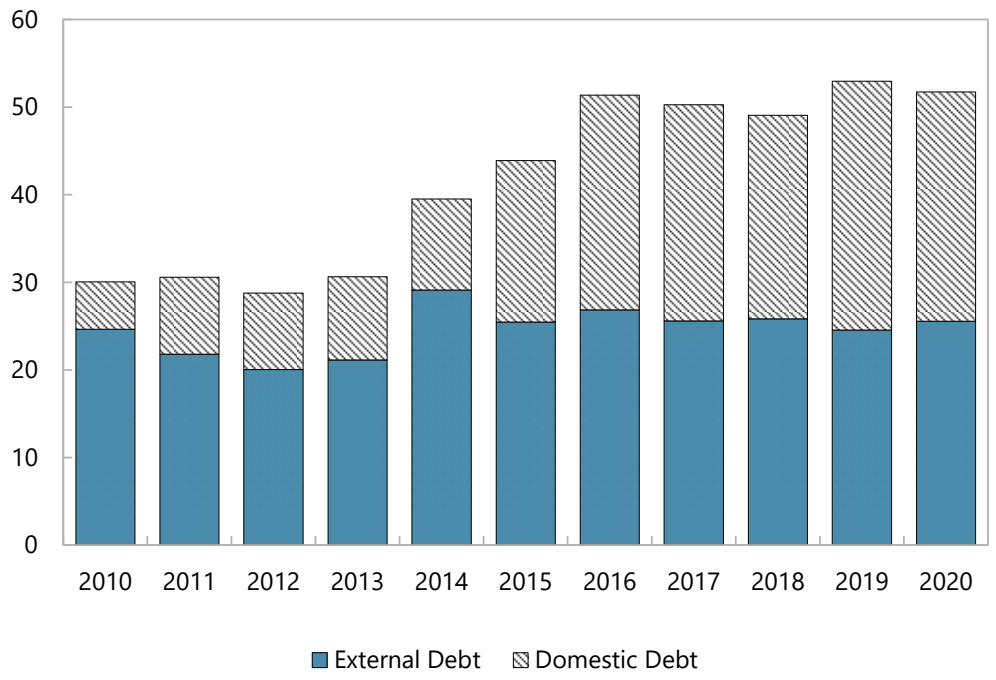
BACKGROUND

A. Evolution and Composition of Debt

1. Chad's external public and publicly guaranteed (PPG) debt burden has stabilized at about 25 percent of GDP since 2015. Chad reached the HIPC Completion Point in April 2015 after a prolonged interim period. In the same year, it also undertook its first rescheduling of oil sale advances from the commercial company, Glencore, to cover revenue shortfalls. The Glencore debt was restructured again in early 2018 (Box 1). At end-2020, outstanding PPG external debt stood at US$3.0 billion, about 26 percent of GDP (Text Table 1).

2. The composition of external PPG debt is dominated by commercial and non-Paris Club lenders. The share of external debt from multilaterals and international creditors has fallen from about 87 percent in 2008, to 36 percent in 2020, which in part reflects grant financing provided by the World Bank (IDA) and the AfDB. In turn, the share of commercial debt, mostly owed to Glencore, increased rapidly following the 2014 crisis. It was reduced following the debt restructuring, from a peak in 2016 of 53 percent to 37 percent at end-2020. Bilateral debt doubled over the past decade with China becoming the main bilateral creditor (Text Table 1).

Text Figure 1. Public and Publicly Guaranteed Debt
(Percent of GDP)



Source: Country authorities, IMF and World Bank staff estimates

Text Table 1. Chad: External Debt Stock 2017–2020[1]
(US$ million, unless otherwise indicated)

	2017	2018	2019	2020
Multilateral	**695**	**741**	**848**	**1071**
IMF	173	262	381	586
World Bank/IDA	182	175	171	170
African Development Bank/ADF	100	95	94	109
Other	239	208	201	206
Official Bilateral	**734**	**800**	**661**	**837**
Paris Club	43	108	122	122
Non-Paris Club	690	692	539	715
China	226	226	218	291
Libya	230	272	230	263
India	38	37	37	48
Other				114
Commercial	**1272**	**1202**	**1158**	**1106**
Glencore Energy	1272	1199	1155	1073
Total	**2702**	**2745**	**2665**	**3013**
Percent of GDP	*25.6*	*25.8*	*24.6*	*26.3*
Memorandum Items				
Multilateral (share of total)	25.7	27.0	31.8	35.5
Bilateral (share of total)	27.2	29.1	24.8	27.8
Commericial (share of total)	47.1	43.8	43.5	36.7

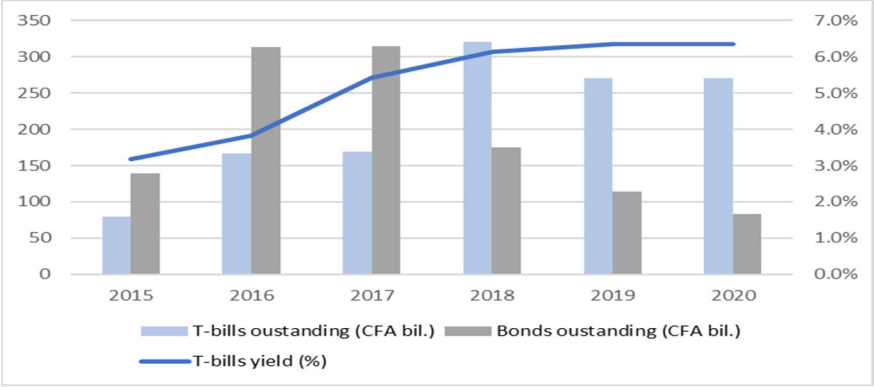
Source: Country authorities, IMF and World Bank staff estimates

3. External payment arrears remain outstanding. As of September 2021, about US$123 million (about 1 percent of GDP) remained outstanding, mainly to bilateral creditors —debt to the Republic of Congo alone is about US$57 million. The authorities are discussing bilaterally arrears with Libya that amount to about US$33 million. Bilateral discussions to restructure the remaining arrears to the Republic of Congo are ongoing. Following a data reconciliation exercise, an agreement on debt restructuring with BDEAC is expected to be finalized soon. After complications related to payment modalities, the authorities reached an agreement in July 2019 to pay the debt owed to Angola in kind (livestock). The authorities have taken steps to prevent further arrears accumulation—including continued arrears clearance during 2020 and 2021, and measures to improve coordination among relevant agencies and enhance debt servicing (e.g., stricter debt service monitoring and reactivation of an escrow account for the payment of external debt at BEAC).

4. Chad has benefited from IDA financial support since 2013 and is also benefiting from debt relief under the CCRT as well as a small amount of debt suspension under the Debt Service Suspension Initiative (DSSI). IDA has significantly stepped up its support since 2013, which has entirely been provided in the form of grants, reflecting IDA's policy to provide grants to countries at moderate and high risk of debt distress. In turn, this has translated into IDA commitments of over $1.5 billion, with positive net flows of over half a billion USD over the past five years. Chad is benefiting from debt relief under the CCRT on its scheduled repayments to the Fund of SDR 10.1 million in 2020 and 2021. Chad has applied for debt suspension under the terms of the G20's DSSI not only from its G20 creditors, but from all of its official bilateral and private commercial creditors. It has signed a MOU with France and Kuwait. Chad has requested its official bilateral creditors to extend the DSSI to end-2021. In 2020, DSSI led to a debt relief of US$1.2 million and is projected to yield US$0.1 million in 2021.

5. An agreement was reached in end-March 2021 to exit the rollover mechanism, which since 2017 had restricted Chad's access to the domestic and regional debt market. To limit the risks from the sovereign-bank relationships in an environment of increased risk perception, a roll-over arrangement was introduced in 2017 by BEAC. This arrangement implied a repayment of 10-15 percent of the marketable debt falling due each year, and the roll-over of the remaining 85-90 percent into 26-week T-bills (BTA). As a result, the profile of the domestic debt is now almost entirely composed of T-bills, exposing Chad to severe refinancing risk (Test Figure 2). An agreement to exit the rollover mechanism was reached in end-March 2021, which allowed Chad to access the regional market, including benefiting from the BEAC bond repurchase program until it expired in September 2021. This mitigated severe liquidity pressures on the treasury during 2021.

Text Figure 2. Domestic Marketable Debt: Issuance Volume (left axis) and Yields (right axis)



Source: BEAC

6. Chad's authorities and BEAC are in the process of reprofiling the existing domestic debt stock. Modalities have not yet been formalized, but the overarching goal is to convert the T-bills (BTA) into long-term bonds (OTA)[2]. Banks' participation in the reprofiling is facilitated by incentives such as the creation of an escrow account for OTA repayments.[3]

7. The authorities have requested debt restructuring under the G20 Common Framework to restore debt sustainability. Despite previous restructuring agreements with Glencore (Box 1), Chad is in debt distress. Chad's elevated debt vulnerabilities are reflected in the persistently high external debt service-to-revenue ratio and the significant financing needs that cannot be realistically met by the existing domestic and external funding sources. The Common Framework debt restructuring aims at addressing these vulnerabilities and restoring a level of external debt stock and service consistent with a moderate risk of external debt distress.

[2] A first five-year bond was successfully auctioned on March 31. A two-year bond was also issued in August 2021. Between April and September, issuances were mostly at a one-year maturity, reflecting lower market appetite following the death of the President in mid-April 2021.

[3] Coupon and principal payments need to be deposited into an escrow account at BEAC at least 10 months before their maturity date. This arrangement reassures investors about Chad's capacity to serve its domestic debt, but it generates a cost of carry, as the escrow account is not remunerated.

Box 1. Glencore's Debt

In June 2014, Chad's National Oil Company (SHT) contracted a loan for US$1.4 billion (around 10 percent of GDP) with Glencore Energy, in turn supported by a consortium of banks, to be repaid over the period 2014-18 through direct deductions from oil shipments sold by Glencore. If the value of deductions from oil shipments turned out to be insufficient to service the debt, then the revenues from oil royalties due to the Government would also be used.

Beginning in early 2015, the Government started negotiating a rescheduling of the agreement with Glencore, since the decline in oil prices that began in mid-2014 implied a sharp decline in oil revenue. In November 2015, a rescheduling agreement was reached, allowing a three-year extension of maturities. This arrangement marginally improved Chad's debt service profile but was not sufficient to place the debt on a sustainable path and Chad's PPG was still classified in debt distress. At end 2016, the debt service to Glencore was absorbing between half and two thirds of the value of total oil shipments, so authorities started seeking a second rescheduling agreement.

In June 2017, the 2014-16 ECF agreed with the IMF was cancelled, and a new ECF-supported program was approved. The authorities decided to achieve a deeper debt restructuring with Glencore[1/], which was concluded in June 2018. The agreement included an extension of maturities, a lower interest rate, and a reduction in restructuring fees that restored external debt sustainability.

The restructured loan includes a mechanism that makes debt service contingent on the Doba oil price, moving above or below certain thresholds[2/]. The upside scenarios generate significant principal payment acceleration and raised interest payments in 2018-2019. Conversely, the downside scenario, which manifested in 2020, involves a limited debt service reduction (1 percent principal deferral). This contingency mechanism also allows Chad to defer some mandatory payments as prices fall, but the cumulative deferred amortization is capped at US$75 million. When the cap is reached, Chad needs to put up cash, potentially necessitating unsustainable fiscal adjustments.

The 2015 and 2018 restructurings involved protracted debt negotiations. The first rescheduling resulted in insufficient debt relief in view of the persistent low oil prices. The second restructuring improved Chad's immediate liquidity constraint but introduced state-contingent features whose parameters limited its impact. These experiences point to the need for a more long-lasting solution, as well as the use of more symmetrical downside and upside parameters, based on independent oil prices projections, should the oil-cash sweep mechanism be maintained in any restructured loan.

[1/]Chad: Request for a Three-Year Arrangement Under the Extended Credit Facility and Cancellation of the Current Arrangement, IMF Country Report No. 17/246, August 2017.

[2/]The complexity of the loan structure hinders a proper accounting by the debt office and an effective cash-management by the national Treasury. The financial accounting related to government oil exports, operating costs and debt service to Glencore is outsourced to an independent third party (Ernst and Young).

PUBLIC DEBT COVERAGE

8. Public debt coverage includes central, state and local governments, as well as government guaranteed external debt owed by the public oil company (*Société des Hydrocarbures du Tchad*, or SHT). Almost all other public sector entities (including other state-owned enterprises (SOEs) and local governments) do not have access to external financing. The exception is the Ndjamena oil refinery (*Société de Raffinage de N'Djaména*, or SRN), in which the central government holds a 40 percent share, and which

has two loans with CNPC Finance and EXIM Bank China. As in previous DSAs, external debt is calculated on a currency basis. Therefore, Chad's CFAF-denominated debt held by the regional development bank (BDEAC),[4] and bilateral creditors in the currency union (Republic of Congo,) are not considered external debt. Debt owed to Angola, which is being repaid in kind and was previously classified as domestic, has been reclassified as external.

9. The contingent liability stress test accounts for vulnerabilities associated with nonguaranteed SOE debt and contingent fiscal liabilities associated with financial sector recapitalization (Text Table 2). Contingent liabilities from financial markets are set at 5 percent of GDP, which represents the average cost to the government of a financial crisis in a low-income country since 1980. The size of the contingent liabilities for the SOE debt is set at 9.5 percent, reflecting the liabilities of SRN, *Société Nationale d'Electricité* (SNE), and *Société Nationale de Ciment* (SONACIM) following the results of a 2019 SOE Census by the World Bank.

Text Table 2. Chad: Coverage of Public-Sector Debt and Design of Contingent Liability Stress Tests

	Subsectors of the public sector	Check box
1	Central government	X
2	State and local government	X
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	

Public debt coverage and the magnitude of the contingent liability tailored stress test

1	The country's coverage of public debt	The central, state, and local governments, central bank, government-guaranteed debt		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	9.5	From the SOE census, 2017 levels.
4	PPP	35 percent of PPP stock	0	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5	
	Total (2+3+4+5) (in percent of GDP)	15		

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

A. Macroeconomic Forecasts

10. The DSA's baseline scenario reflects policies and financing assumptions consistent with the requested ECF arrangement. Relative to the previous DSA (August 2020), real GDP growth has been reduced from 6.1 percent in 2021 to 0.6 percent. Both oil and non-oil GDP are expected to modestly rebound in 2021, although less strongly than in the previous projections since oil producers have adjusted

4 The *Banque de Développement des Etats de l'Afrique Centrale* (BDEAC) is the development bank of the CEMAC region. The main shareholder is BEAC (33.43 percent). CEMAC countries, including Chad, are equal shareholders (8.48 percent).

down their oil output projections. Oil production is expected to resume growth in the medium term, leading to higher oil revenues, exports, and overall GDP growth. The outlook assumes a revenue-led fiscal consolidation that will continue beyond the program horizon at a gradual pace and that spending control would also be maintained. The fiscal consolidation would aim to maintain public investment at reasonable levels and protect social spending while improving its efficiency. Chad ranks second to last in the World Bank's Human Capital Index 2020 and about 6.5 million Chadians or 42 percent of the population are poor, according to national statistics, and it is highly vulnerable to climate change.[5] At the same time, social spending and spending on climate adaptation are among the lowest in the world. Any fiscal consolidation should therefore not only protect but allow for increase spending for critical social sectors and climate adaptation.

11. The forecast is subject to heightened uncertainty and downside risks as the economic impact of the pandemic and ongoing insecurity unfolds, and as Chad is subject to frequent climatic shocks. Chad is one of the poorest and most vulnerable countries in the world, and its heavy dependency on oil has increased economic volatility as witnessed by four recessions since 2006, with significant consequences for debt sustainability. Insecurity and increasing climatic shocks exacerbate these downside risks.[6] The baseline scenario assumes a gradual clearance of external arrears on debt over the period 2021-26. The authorities have a financing plan that should underpin a gradual repayment of audited domestic arrears. Chad intends also to resume the clearing of outstanding audited domestic arrears (equivalent to approximately 8 percent of the GDP) of which CFA 235 billion (3.6 percent of GDP) are expected to be cleared during 2021-24.

12. Over the long term, the DSA assumes an average real growth rate of 3.8 percent for 2025-29. While this is higher than the recent growth performance for Chad, it is grounded in the reform agenda on which the government is committed to embark on in the coming years. The government is already taking steps to mobilize more non-oil revenues, including digitizing customs procedures for livestock and streamlining tax exemptions. Improving fiscal space will support an increase in public investment to average around 7 percent of GDP. A higher level, and better governance, of public investment in energy and connectivity infrastructure coupled with reforms to improve the business environment will unleash the potential of the private non-oil sector in Chad and sustain a recovery. In the long run, the DSA is also projecting a GDP deflator of 2.9 percent, consistent with the BEAC monetary policy objective and historical data.

13. Financing assumptions have been updated based on most recent information. This DSA corresponds to a pre-restructuring scenario. With respect to external financing, the DSA includes IDA19 grants consistent with Chad's performance-based allocation that averages around $140 million between 2021 and 2022 (Text Figure 3).[7] Subsequently, annual IDA allocations assume a similar level of resources as the IDA19 performance-based allocation. Actual IDA financing would be dependent on the performance of Chad's reforms and subsequent replenishments by IDA Deputies, with terms based on current IDA

[5] World Bank. 2020. The Next Generation Africa Climate Business Plan: Ramping Up Development-Centered Climate Action. World Bank, Washington, DC.

[6] The most recent development with an armed rebellion and a military-led transition government further highlights the volatility of the security situation.

[7] For FY2021, an additional $75 million was committed through the IDA Refugee Window for Chad.

policies[8]. A significant portion of the IDA resources will be deployed to support social investment projects to reduce fragility. New financing is also assumed starting in 2023 from the AfDB, and other international and bilateral partners, which is conditional on a successful debt restructuring. Overall, given the authorities' limited capacity to borrow on the regional debt market, especially after the April 2021 events, their debt strategy mainly relies on concessional external financing and grants to meet their financing needs.

Text Table 3. Chad: Selected Macroeconomic Assumptions[1]

	2017-20	2021	2022	2023	2024	2025-29 (avg.)
	(Annual percentage change, unless otherwise indicated)					
Real GDP growth						
Current DSA	0.3	0.6	2.2	3.1	3.6	3.8
August 2020 DSA	0.5	6.1	4.9	4.0	3.8	3.8
Oil GDP						
Current DSA	1.9	2.3	2.3	2.0	2.4	2.9
August 2020 DSA	2.9	19.2	10.6	4.4	2.8	2.9
Non-oil GDP						
Current DSA	0.0	0.2	2.1	3.3	3.9	4.1
August 2020 DSA	0.0	3.1	3.4	3.8	4.1	4.1
Current Account Balance (incl. official Transfers) (% of GDP)						
Current DSA	-5.4	-6.5	-5.8	-7.3	-7.9	-5.4
August 2020 DSA	-6.7	-10.8	-8.8	-8.7	-8.4	-3.4
	(percent of non-oil GDP, unless otherwise indicated)					
Overall budget balance2						
Current DSA	0.5	-3.4	0.8	0.2	0.0	0.9
August 2020 DSA	0.4	-3.8	-1.3	1.0	0.4	2.6
Revenue and grants						
Current DSA	19.2	19.2	22.0	21.3	20.7	20.4
August 2020 DSA	19.4	17.6	19.4	21.2	20.5	21.7
of which: oil revenues						
Current DSA	7.0	7.0	9.1	7.8	6.7	5.6
August 2020 DSA	7.0	5.0	6.0	7.4	6.0	6.7
of which: non-oil revenue						
Current DSA	8.8	9.3	9.8	10.3	10.8	11.9
August 2020 DSA	8.7	8.5	9.4	9.9	10.6	11.7
of which: grants						
Current DSA	3.4	3.0	3.0	3.2	3.2	3.0
August 2020 DSA	3.7	4.1	4.5	4.9	5.1	5.3
Grant Element of new External Borrowing (Current DSA, percent)	....	35.4	39.4	40.2	40.9	42.4

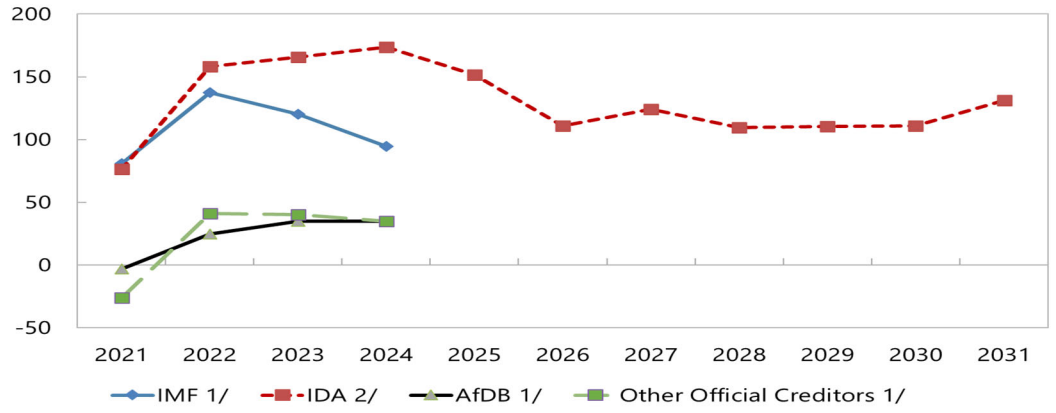
Source: Country authorities, IMF and World Bank staff estimates

1/ DSA Vintage used for RCF-2 Arrangment.

2/ Commitment basis, including grants.

[8] Consistent with the LIC DSF Guidance Note, for the World Bank (IDA) and other MDBs, regular credit terms on all lending are assumed for all years in the projection period for which grant finance has not already been committed.

Text Figure 3. Chad: Official External Financing, Net
($US, millions)



Sources: IMF and World Bank Staff estimates
1/ Financing commitments currently unavailable beyond 2024
2/ Includes project grants

14. The fiscal adjustment envisaged is realistic and relies on a combination of increases in non-oil revenue and streamlining non-essential expenditures. The projected four-year fiscal adjustment in the non-oil primary balance is ambitious compared with other LIC adjustment programs but is consistent with Chad's previous adjustment between 2014-17.[9]Over the 36 months, the program targets a consolidation of the NOPB of 4.0 percentage points of non-oil GDP through the mobilization of non-oil revenues and streamlining expenditures, while promoting social spending. Tax revenues are targeted to increase by 1.9 percentage points of non-oil GDP. Current expenditures would drop by 2.0 percentage points of non-oil GDP by gradually reducing the wage bill and subsidies to the electricity company. Capital expenditures are expected to remain at an average of 7.3 percent of non-oil GDP, but above pre-pandemic levels. The rationalization of public spending, mainly through strengthening expenditure controls, prudent increases and efficiency gains in social spending, and improvements in public investment efficiency, combined with greater transparency and oversight over debt management, will be required to reduce the risk of future debt distress.

15. The DSA assumptions incorporate a gradual increase in social spending (34 percent of current spending over 2021-24) within the overall fiscal consolidation.[10] Specifically, the framework incorporates the government's commitments to recruit additional staff in the health and education sectors, who will be deployed to underserved regions, while recruitments on other areas will be closely monitored. The framework enables public capital expenditure to rise from a relatively modest level, which should be beneficial for growth and would enable Chad to address large infrastructure gaps. Ensuring greater

[9] This assessment is different from the one shown in Figure 5 below, which refers to the adjustment in the primary fiscal balance rather than in the non-oil primary balance as a share of non-oil GDP. The latter is the relevant metric for an oil-exporting country such as Chad.

[10] Poverty–reducing social spending, according to the latest general structure of Government , comprises public spending by the following ministries: (i) National Education and Civic Promotion; (ii) Public Health, including military health services and National Solidarity; (iii) Women, Early Childhood Protection and National Solidarity; (iv) Production, Irrigation and Agricultural Equipment; (v) Livestock and Animal Production; (vi) Environment Water and Sanitation; (vii) Professional Training and Small Job Promotion; and (viii) Higher Education.

transparency and strengthening procedures and processes in critical sectors, such as health, education, energy, agriculture, transport, and public investment management, will help to promote better economic and social outcomes.

B. Country Classification and Determination of Stress Test Scenarios

16. The composite indicator (CI) based on October 2020 World Economic Outlook (WEO) projections and an update of the CPIA index to 2020 levels indicates weak debt carrying capacity for Chad. The CI combines the CPIA score, external conditions as captured by world economic growth, and country-specific factors. The October 2020 data indicate weak debt carrying capacity, reflecting mainly a low CPIA, very low remittances, and a low level of foreign reserves (Text Table 4). Debt-carrying capacity was weak before the latest update.

17. A tailored commodity price stress test is activated for Chad. Debt service under the Glencore contract includes a mandatory amortization and interest payment plus a cash-sweep component that falls as the Doba oil price goes below a threshold. Because oil prices have fallen so far in the baseline, this contingency is fully exercised even in the baseline and the standard commodity price shock scenarios. As a second contingency mechanism, the contract allows Chad to defer some mandatory payments as prices fall, but the cumulative deferred amortization is capped at $75 million. For this reason, the customized scenario featured in the last DSA is no longer informative.

Text Table 4. Chad: CI Score

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.38	2.72	1.05	42%
Real growth rate (in percent)	2.72	2.51	0.07	3%
Import coverage of reserves (in percent)	4.05	31.30	1.27	51%
Import coverage of reserves^2 (in percent)	-3.99	9.80	-0.39	-16%
Remittances (in percent)	2.02	0.00	0.00	0%
World economic growth (in percent)	13.52	3.50	0.47	19%
CI Score			2.47	100%
CI rating			Weak	

Source: IMF and WB staff calculations. The CI cutoff for medium debt carrying capacity is 2.69.

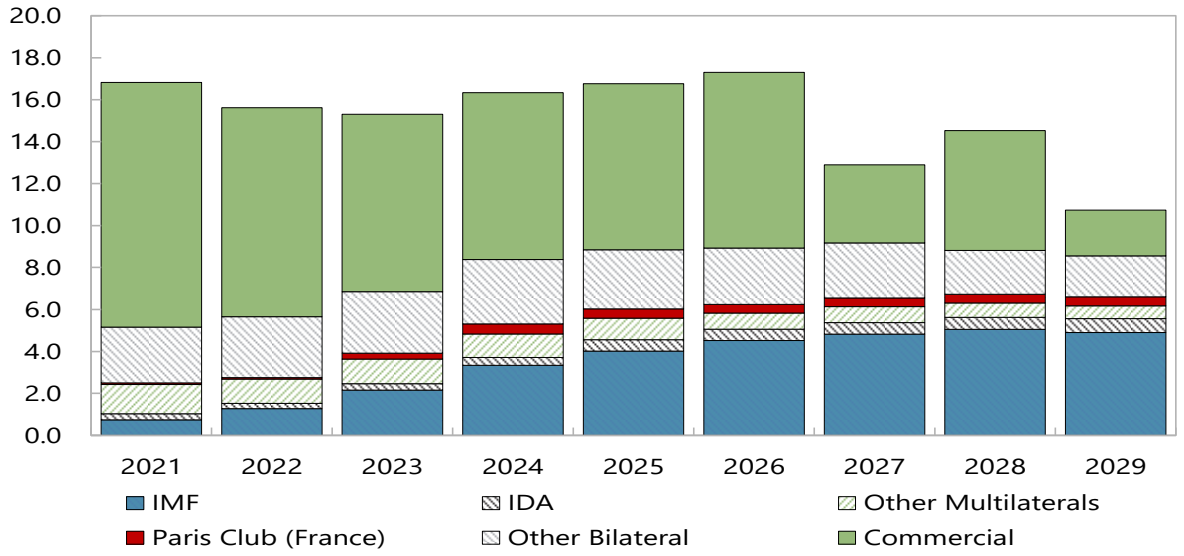
DEBT SUSTAINABILITY

A. External Debt Sustainability

18. The external debt service-to-revenue ratio threshold under the baseline scenario is breached. The debt service-to-revenue ratio rises above 14 percent starting in 2021 due to lower revenue and associated higher borrowing in response to the pandemic. This ratio is not expected to permanently drop below its threshold of 14 percent until after 2029, as the Glencore debt matures (Text Figure 4). Under the baseline scenario, the other liquidity and solvency ratios remain below their thresholds (Figure 4).

19. The thresholds for all indicators are breached under stress tests. Under the shock scenarios, the exports stress test produces the most extreme scenario for all indicators except the debt service-to-revenue ratio, for which the commodity price stress test is the most extreme.[11] Under the exports stress test, all four indicator thresholds are breached for the entire horizon (Figure 2). For the present value of PPG external debt-to GDP, present value of PPG external debt-to-exports, and debt service-to-exports ratios, levels approach those seen during the 2018 debt distress episode. The debt service-to-revenue-ratio—the factor that pushed Chad into debt distress in the past—peaks at 22 percent.

Text Figure 4. Chad: External Debt Service by Creditor
(percent of government revenue, excluding grants)



Source: Country authorities, IMF and World Bank staff estimates.
In line with the LIC DSF Guidance Note, future IDA disbursements are assumed in the form of concessional loans.

B. Public Debt Sustainability

20. The benchmark for public debt is breached in 2021-26 under the baseline. Due to higher budget deficits related to the impact of the COVID-19 pandemic, the PV of total public debt-to-GDP ratio is projected to peak at 45 percent in 2022, above the 35 percent high-risk threshold associated with heightened public debt vulnerabilities with a weak debt carrying capacity. The benchmark for public debt is also breached through 2030 for the most extreme contingency scenario. The DSA assumes a normalization of the domestic issuance capacity which would partially fill the financing gap.

C. Risk Rating and Vulnerabilities/Staff's Assessment

21. Chad is in debt distress. Based on the assessment of the external public debt indicators and ongoing debt restructuring, Chad's classification has been changed to "in debt distress" from high risk of debt distress in the previous DSA (August 2020). Given the impact of COVID and other shocks on growth and the widening of the fiscal and external imbalances, there is a large residual financing need that cannot be met with fiscal adjustment and grants alone. Moreover, Chad's outlook is subject to significant

[11] A tailored stress test, tailored to reflect a 20 percent reduction in oil price, is activated for Chad.

downside risk, related to the economic impact of the pandemic, ongoing insecurity, and exposure to climatic shocks.

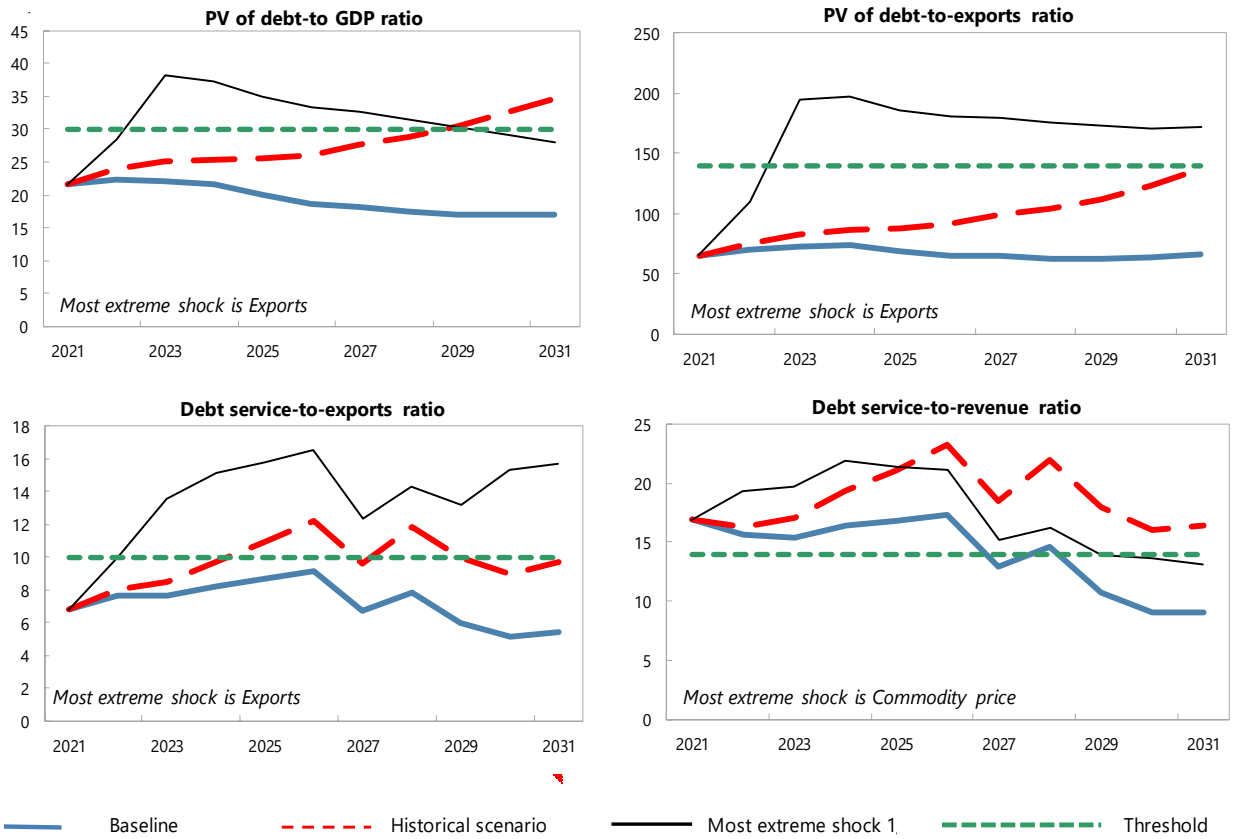
22. A deep debt restructuring under the Common Framework is underway and is expected to restore debt sustainability and bring the risk rating to 'moderate' within the program period. Given the request for exceptional access under the proposed Fund arrangement and the current 'in debt distress' rating, Fund policies require that the envisaged debt restructuring should bring the risk of external debt distress back to a low or moderate level. The envelope for the restructured debt will need to allow for a suitable buffer to absorb shocks as the economy is vulnerable to frequent and often concurrent shocks. For example, recent economic shocks have resulted from global oil price volatility, regional and domestic insecurity (with Chad hosting large numbers of internally displaced people and refugees), climatic shocks with droughts and flooding of increasing intensity, in addition to the COVID-19 pandemic. Given the liquidity challenges, the debt restructuring envelope will need to close the 2022-24 residual financing gap. The Chadian authorities have received financing assurances from the Common Framework creditor committee for a restructuring in line with these parameters and have also taken important steps towards the required restructuring of debt held by private creditors.

23. Following debt restructuring, significant concessional financing will still be required to maintain a moderate risk of external and overall debt distress. While restructuring is expected to reduce the annual financing gap, significant concessional financing commitments by multilateral and bilateral creditors will still be required over the longer-term to avoid a recurrence of either external and/or domestic debt service problems. A zero NCB limit is an important part of the debt conditionality under the proposed Fund-supported program. The government would also need to take action to further enhance debt management capacity and public debt transparency while strengthening fiscal policies. In particular, improving control of SOE liabilities, enhancing domestic revenue mobilization and expenditure efficiency, and normalizing relations in the regional debt market. These policy measures are important since adverse shocks to revenues (i.e. from oil price shocks, insecurity) and climatic conditions, are likely to be a persistent feature in Chad over the long term.

D. Authorities' Views

24. The authorities broadly agreed with the overall assessment of the country's debt sustainability. Debt sustainability is expected to be restored through a deep debt restructuring, donor support, and the reforms contemplated under the Fund-supported program. The authorities are committed to further improve debt management.

Figure 1. Chad, Indicators of Public and Publicly Guaranteed External Debt under Restructuring Scenarios, 2021-31



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	Yes	
Natural Disasters	n.a.	n.a.
Commodity Prices 2/	No	No
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	0.8%	0.8%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	21	20
Avg. grace period	7	6

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

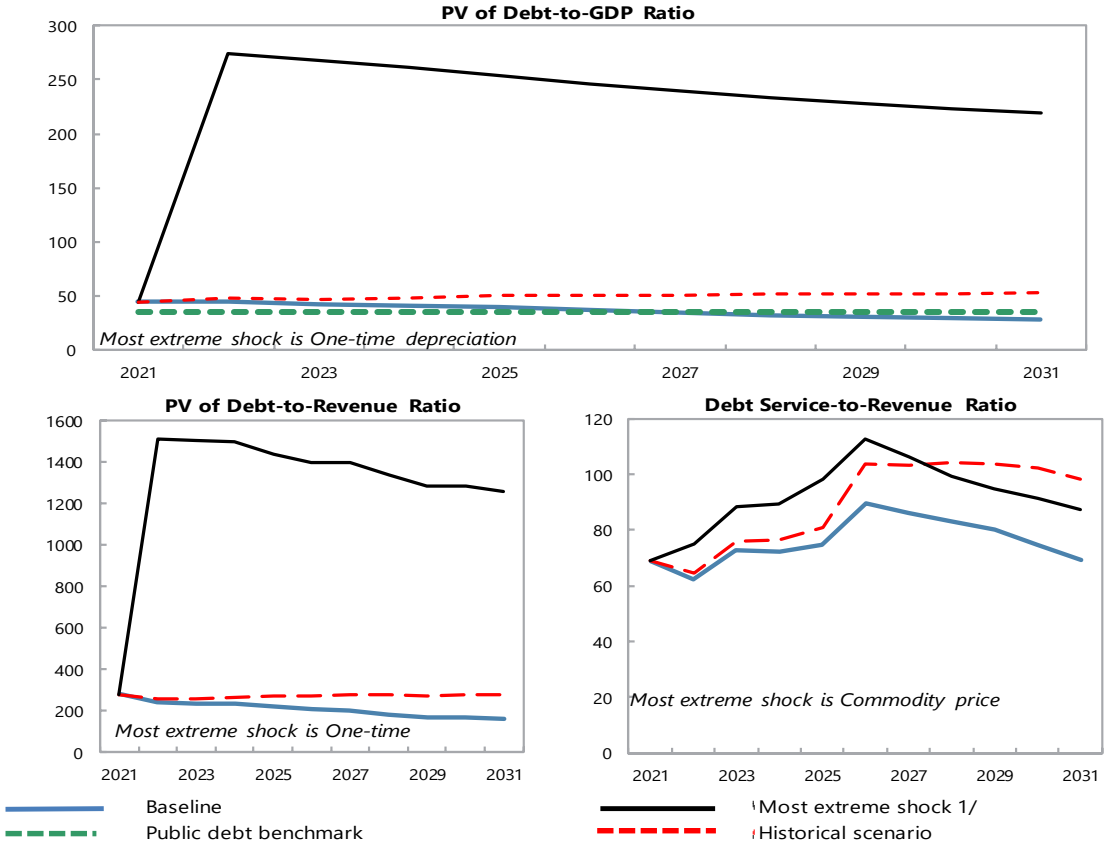
Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

The tailored commodity price stress test presented here does not account for the contingency mechanisms in the Glencore debt as Text Figure 1 does.

Figure 2. Chad: Indicators of Public Debt Under Alternative Scenarios, 2021-2031



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	37%	65%
Domestic medium and long-term	10%	5%
Domestic short-term	115%	30%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	0.8%	0.8%
Avg. maturity (incl. grace period)	21	20
Avg. grace period	7	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	5.2%	5.0%
Avg. maturity (incl. grace period)	6	1
Avg. grace period	0	0
Domestic short-term debt		
Avg. real interest rate	0%	2.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

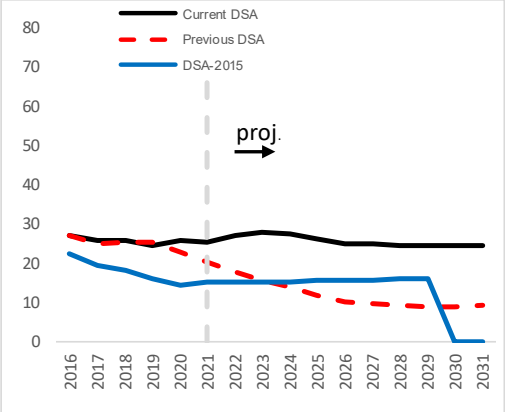
Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

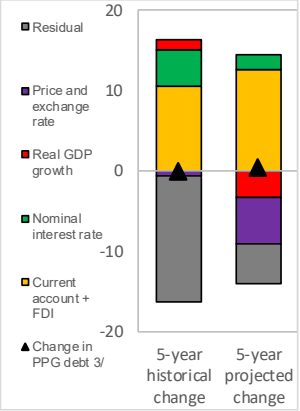
Figure 3. Chad: Drivers of Debt Dynamics- Baseline Scenario

External debt

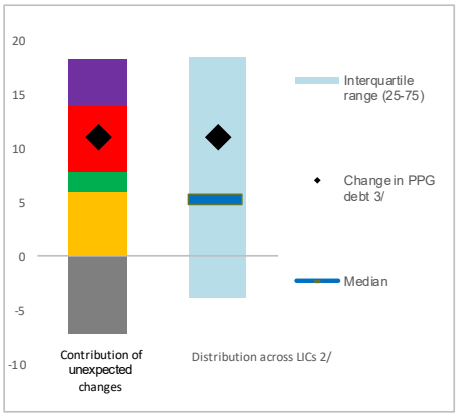
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

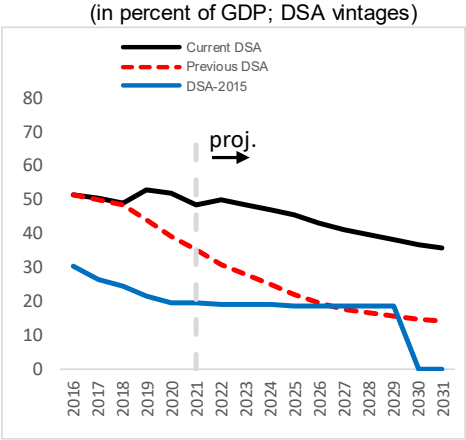


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

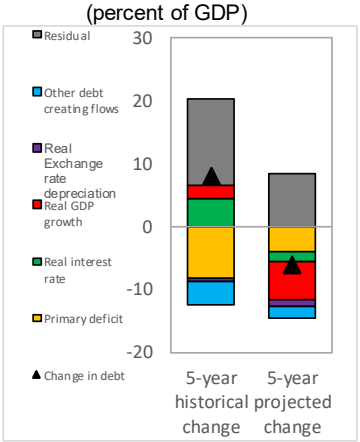


Public debt

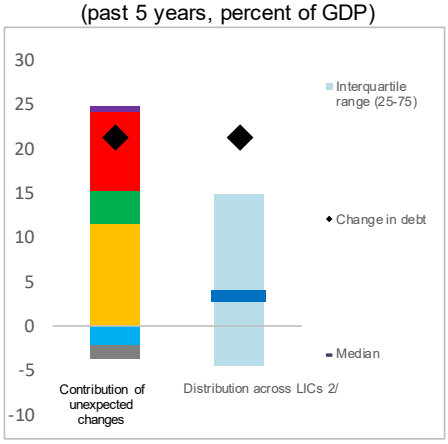
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



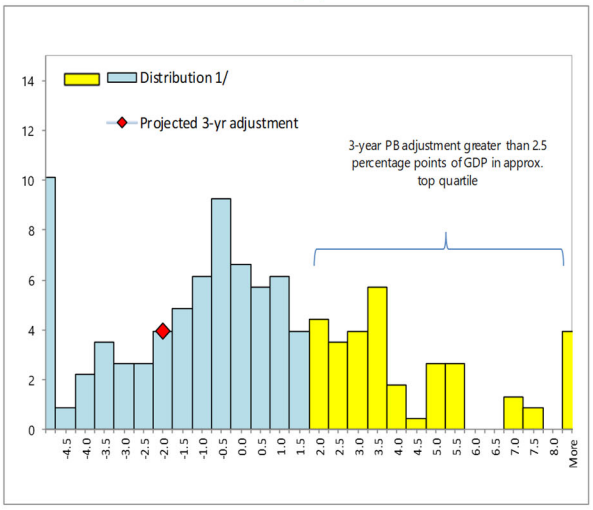
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

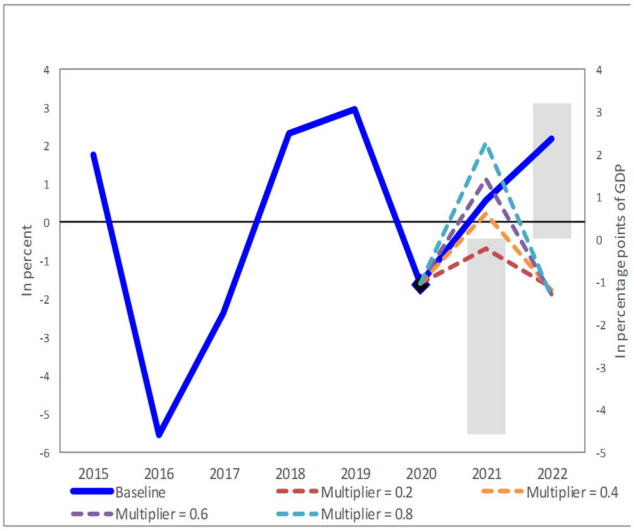
Figure 4. Chad: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



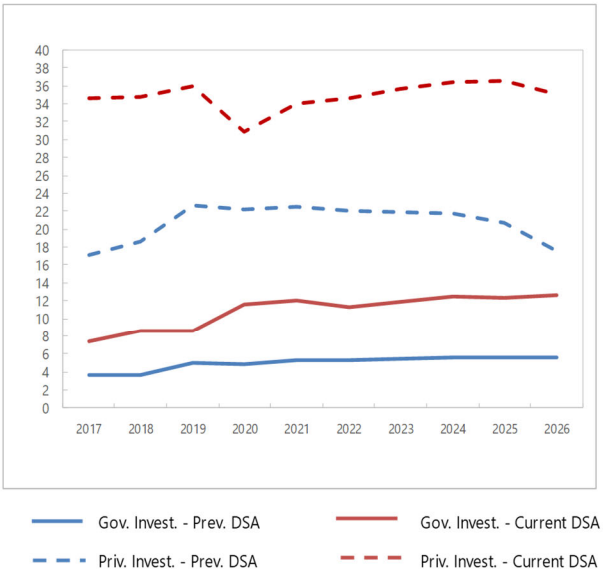
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/

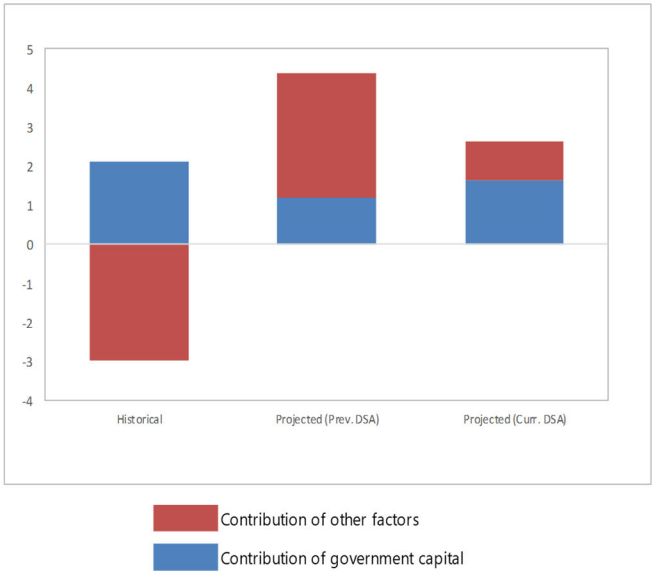


1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(% of GDP)



Contribution to Real GDP growth
(percent, 5-year average)



Source: Country authorities, IMF and World Bank staff estimates

Table 1. Chad: External Debt Sustainability Framework, Baseline Scenario, 2010–2041

	Actual											Projections								Average 8/	
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projectio
External debt (nominal) 1/	**24.6**	**21.8**	**20.0**	**21.1**	**29.1**	**25.5**	**26.8**	**25.6**	**25.8**	**24.6**	**25.5**	**25.4**	**27.0**	**27.7**	**27.6**	**26.0**	**24.9**	**24.4**	**19.8**	**24.6**	**25.5**
of which: public and publicly guaranteed (PPG)	**24.6**	**21.8**	**20.0**	**21.1**	**29.1**	**25.5**	**26.8**	**25.6**	**25.8**	**24.6**	**25.5**	**25.4**	**27.0**	**27.7**	**27.6**	**26.0**	**24.9**	**24.4**	**19.8**	**24.6**	**25.5**
Change in external debt	...	-2.8	-1.7	1.1	8.0	-3.7	1.4	-1.3	0.2	-1.3	1.0	-0.2	1.6	0.7	-0.1	-1.6	-1.1	0.0	-0.8		
Identified net debt-creating flows	**...**	**-1.7**	**2.7**	**4.2**	**2.2**	**16.8**	**9.9**	**3.8**	**-3.8**	**0.7**	**5.3**	**2.8**	**1.5**	**2.6**	**3.0**	**1.3**	**1.3**	**0.8**	**-0.1**	**4.0**	**1.5**
Non-interest current account deficit	**8.4**	**5.4**	**7.1**	**8.5**	**8.2**	**12.6**	**8.7**	**6.1**	**0.8**	**4.1**	**7.9**	**6.0**	**5.4**	**6.9**	**7.6**	**6.0**	**5.6**	**1.8**	**0.9**	**6.9**	**5.1**
Deficit in balance of goods and services	10.8	7.5	9.8	9.6	12.5	16.5	16.2	15.2	8.7	9.6	17.2	11.8	12.4	12.9	12.3	10.8	10.1	6.9	5.0	**12.3**	**10.2**
Exports	37.8	40.6	38.2	33.4	31.5	26.4	22.9	26.5	33.3	34.8	27.3	32.9	31.7	30.3	29.3	29.2	28.5	25.3	18.7		
Imports	48.6	48.1	48.0	43.1	43.9	42.9	39.1	41.7	42.0	44.3	44.5	44.7	44.1	43.3	41.6	40.0	38.6	32.3	23.7		
Net current transfers (negative = inflow)	-5.6	-4.3	-4.4	-5.1	-7.9	-7.1	-7.6	-9.4	-8.9	-7.4	-10.9	-7.8	-9.0	-8.1	-6.8	-6.6	-6.3	-5.4	-3.9	**-7.3**	**-6.6**
of which: official	-0.3	-0.3	-0.7	-1.3	-4.0	-2.5	-2.4	-3.1	-3.2	-1.3	-3.3	-1.5	-1.6	-1.6	-1.5	-1.5	-1.5	-1.3	-1.2		
Other current account flows (negative = net inflow)	3.2	2.2	1.7	3.9	3.6	3.2	0.1	0.3	0.9	2.0	1.6	2.0	1.9	2.1	2.0	1.7	1.8	0.2	-0.1	**2.0**	**1.6**
Net FDI (negative = inflow)	**-5.2**	**-4.5**	**-4.7**	**-4.0**	**-5.2**	**-5.1**	**-2.4**	**-3.6**	**-3.0**	**-4.3**	**-3.6**	**-3.5**	**-3.8**	**-4.0**	**-4.0**	**-4.1**	**-3.7**	**-0.6**	**-0.6**	**-4.0**	**-3.3**
Endogenous debt dynamics 2/	**...**	**-2.6**	**0.3**	**-0.3**	**-0.8**	**9.3**	**3.6**	**1.4**	**-1.6**	**0.9**	**1.0**	**0.3**	**-0.1**	**-0.4**	**-0.6**	**-0.7**	**-0.6**	**-0.4**	**-0.4**		
Contribution from nominal interest rate	...	0.4	0.7	0.6	0.7	1.1	1.7	1.0	0.6	0.7	0.6	0.4	0.4	0.4	0.4	0.3	0.3	0.2	0.2		
Contribution from real GDP growth	...	0.0	-1.9	-1.1	-1.4	-0.7	1.5	0.6	-0.5	-0.8	0.4	-0.1	-0.5	-0.8	-0.9	-1.0	-0.9	-0.6	-0.6		
Contribution from price and exchange rate changes	...	-3.0	1.5	0.2	-0.2	8.8	0.4	-0.3	-1.7	1.0	0.1	...	...	...	...	...	...	...	...		
Residual 3/	**...**	**-1.1**	**-4.5**	**-3.2**	**5.8**	**-20.4**	**-8.5**	**-5.1**	**4.1**	**-2.0**	**-4.3**	**-3.0**	**0.1**	**-1.9**	**-3.1**	**-2.8**	**-2.4**	**-0.8**	**-0.7**	**-3.9**	**-1.6**
of which: exceptional financing	...	0.0	0.0	0.0	-0.1	-0.8	-1.1	-1.0	-1.3	-1.6	-0.4	-1.0	-1.7	-1.6	-1.5	-0.3	-0.3	-0.2	-0.1		
Sustainability indicators																					
PV of PPG external debt-to-GDP ratio	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**23.5**	**21.6**	**22.2**	**22.2**	**21.5**	**19.9**	**18.7**	**16.9**	**13.8**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**...**	**86.0**	**65.6**	**70.1**	**73.0**	**73.5**	**68.4**	**65.6**	**66.7**	**73.8**		
PPG debt service-to-exports ratio	**1.5**	**2.2**	**3.2**	**3.9**	**15.6**	**9.6**	**14.4**	**9.0**	**6.1**	**4.1**	**6.5**	**6.8**	**7.7**	**7.6**	**8.2**	**8.7**	**9.1**	**5.4**	**6.6**		
PPG debt service-to-revenue ratio	**3.1**	**3.9**	**5.7**	**7.1**	**31.0**	**24.0**	**34.5**	**22.4**	**16.9**	**11.3**	**10.4**	**16.9**	**15.7**	**15.4**	**16.4**	**16.8**	**17.4**	**9.1**	**9.2**		
Gross external financing need (Million of U.S. dollars)	...	...	...	...	1114.2	1101.2	980.5	486.7	-24.7	132.0	647.5	568.4	508.8	709.5	858.8	686.0	746.0	565.0	623.3		
Key macroeconomic assumptions																					
Real GDP growth (in percent)	13.6	0.1	8.8	5.8	6.9	1.8	-5.6	-2.4	2.3	3.0	-1.6	0.6	2.2	3.1	3.6	3.8	3.7	2.6	2.9	**1.9**	**3.1**
GDP deflator in US dollar terms (change in percent)	1.1	13.7	-6.4	-1.0	0.8	-23.2	-1.4	1.2	7.0	-3.8	-0.2	11.2	2.5	3.3	3.4	3.5	3.3	3.1	0.3	**-1.3**	**3.8**
Effective interest rate (percent) 4/	...	1.8	3.3	3.4	3.7	3.1	6.2	3.8	2.8	2.6	2.2	1.9	1.8	1.5	1.4	1.3	1.1	0.9	0.9	**3.3**	**1.5**
Growth of exports of G&S (US dollar terms, in percent)	23.2	22.3	-4.1	-8.4	1.4	-34.4	-19.2	14.5	37.6	3.3	-22.9	34.8	1.0	1.8	3.5	6.9	4.7	1.0	7.9	**-1.0**	**6.5**
Growth of imports of G&S (US dollar terms, in percent)	19.4	12.7	1.6	-6.0	9.9	-23.7	-15.1	5.4	10.3	4.5	-1.3	12.2	3.3	4.4	3.0	3.3	3.3	-1.4	5.5	**-0.2**	**3.9**
Grant element of new public sector borrowing (in percent)		...	...	...	...	...	...	...	...	...	...	35.4	39.4	40.2	40.9	43.4	42.2	42.2	47.8	**...**	**41.1**
Government revenues (excluding grants, in percent of GDP)	18.9	23.2	21.7	18.5	15.8	10.5	9.5	10.6	12.0	12.6	16.9	13.2	15.5	15.1	14.7	15.0	15.0	15.0	13.4	**15.1**	**14.8**
Aid flows (in Million of US dollars) 5/	263.4	279.6	477.1	377.5	369.1	433.3	326.8	527.7	511.9	283.6	577.9	521.0	789.9	766.0	765.0	771.6	904.9	1084.0	985.2		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	...	...	...	...	...	...	...	...	3.7	4.8	4.7	4.5	4.0	4.1	3.7	2.2	**...**	**4.1**
Grant-equivalent financing (in percent of external financing)	...	...	...	...	...	...	...	...	...	...	...	67.9	61.4	64.8	66.9	73.8	70.3	72.4	84.0	**...**	**69.3**
Nominal GDP (Million of US dollars)	10,701	12,183	12,411	12,994	14,003	10,952	10,202	10,079	11,036	10,934	10,735	12,004	12,567	13,378	14,331	15,395	16,501	22,395	40,167		
Nominal dollar GDP growth	...	13.8	1.9	4.7	7.8	-21.8	-6.8	-1.2	9.5	-0.9	-1.8	11.8	4.7	6.5	7.1	7.4	7.2	5.8	3.2	**0.5**	**6.9**
Memorandum items:																					
PV of external debt 7/	...	...	...	...	...	...	...	...	...	...	23.5	21.6	22.2	22.2	21.5	19.9	18.7	16.9	13.8		
In percent of exports	...	...	...	...	...	...	...	...	...	...	86.0	65.6	70.1	73.0	73.5	68.4	65.6	66.7	73.8		
Total external debt service-to-exports ratio	1.5	2.2	3.2	3.9	15.6	9.6	14.4	9.0	6.1	4.1	6.5	6.8	7.7	7.6	8.2	8.7	9.1	5.4	6.6		
PV of PPG external debt (in Million of US dollars)											2520.5	2591.4	2795.7	2965.4	3088.2	3070.9	3082.7	3779.9	5546.5		
(PVt-PVt-1)/GDPt-1 (in percent)												0.7	1.7	1.4	0.9	-0.1	0.1	0.9	-0.1		
Non-interest current account deficit that stabilizes debt ratic	...	8.3	8.8	7.4	0.2	16.3	7.4	7.3	0.5	5.4	6.9	6.2	3.8	6.2	7.7	7.6	6.6	1.8	1.7		

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 2. Chad: Public Sector Debt Sustainability Framework, Baseline Scenario, 2010–2041

	Actual											Projections								Average 7/	
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
Public sector debt 1/	**24.6**	**30.6**	**28.8**	**30.6**	**39.5**	**43.9**	**51.3**	**50.3**	**49.1**	**53.0**	**51.7**	**48.4**	**49.9**	**48.2**	**47.0**	**45.6**	**43.0**	**35.7**	**25.2**	**42.9**	**43.0**
of which: external debt	**24.6**	**21.8**	**20.0**	**21.1**	**29.1**	**25.5**	**26.8**	**25.6**	**25.8**	**24.6**	**25.5**	**25.4**	**27.0**	**27.7**	**27.6**	**26.0**	**24.9**	**24.4**	**19.8**	**24.6**	**25.5**
Change in public sector debt	...	5.9	-1.8	1.9	8.9	4.4	7.5	-1.1	-1.2	3.9	-1.2	-3.3	1.5	-1.7	-1.1	-1.5	-2.6	-1.0	-0.3		
Identified debt-creating flows		**-4.5**	**-3.0**	**0.6**	**0.2**	**10.2**	**5.4**	**-3.1**	**-3.8**	**-1.8**	**-2.7**	**-1.0**	**-3.2**	**-3.1**	**-3.2**	**-3.9**	**-4.0**	**-2.2**	**-0.3**	**-0.2**	**-3.0**
Primary deficit /2	**3.6**	**-3.0**	**-0.9**	**1.5**	**3.6**	**2.7**	**-0.1**	**-1.3**	**-3.0**	**-0.8**	**-3.0**	**1.5**	**-1.7**	**-1.1**	**-1.0**	**-1.6**	**-1.8**	**-0.9**	**0.1**	**-0.4**	**-1.1**
Revenue and grants	20.2	24.8	24.4	20.7	17.8	14.0	12.4	14.6	15.3	14.2	21.1	15.9	18.4	18.0	17.7	17.9	17.8	17.7	15.3	**17.9**	**17.6**
of which: grants	1.3	1.6	2.6	2.3	2.0	3.4	2.9	4.0	3.3	1.7	4.2	2.8	2.8	3.0	3.0	2.9	2.9	2.7	1.8		
Primary (noninterest) expenditure	23.8	21.8	23.4	22.3	21.4	16.7	12.4	13.3	12.3	13.4	18.1	17.5	16.7	16.9	16.7	16.3	16.0	16.7	15.4	**17.5**	**16.5**
Automatic debt dynamics		**-1.5**	**-2.1**	**-0.4**	**1.0**	**8.2**	**5.9**	**0.0**	**-0.2**	**-0.5**	**0.8**	**-2.2**	**-1.2**	**-1.6**	**-1.9**	**-2.0**	**-1.9**	**-1.1**	**-0.3**		
Contribution from interest rate/growth differential	...	-0.1	-2.3	-0.8	-1.5	1.1	4.4	2.4	-1.2	-1.2	2.2	-2.0	-1.0	-1.5	-1.7	-1.7	-1.6	-0.9	-0.3		
of which: contribution from average real interest rate	...	-0.1	0.2	0.8	0.5	1.7	1.8	1.2	-0.1	0.2	1.3	-1.7	0.0	0.0	0.0	0.0	0.0	0.1	0.4		
of which: contribution from real GDP growth	...	0.0	-2.5	-1.6	-2.0	-0.7	2.6	1.3	-1.1	-1.4	0.9	-0.3	-1.0	-1.5	-1.7	-1.7	-1.6	-0.9	-0.7		
Contribution from real exchange rate depreciation		-1.4	0.2	0.3	2.5	7.1	1.5	-2.4	1.0	0.7	-1.4	...	...	...	...	...	...	...	...		
Other identified debt-creating flows		**0.0**	**-0.1**	**-0.5**	**-4.4**	**-0.7**	**-0.5**	**-1.8**	**-0.6**	**-0.4**	**-0.4**	**-0.4**	**-0.4**	**-0.3**	**-0.3**	**-0.3**	**-0.3**	**-0.2**	**-0.1**	**-0.9**	**-0.3**
Privatization receipts (negative)		0.0	-0.1	0.0	-4.0	-0.4	0.0	-1.2	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)		0.0	0.0	-0.5	-0.4	-0.3	-0.5	-0.5	-0.4	-0.4	-0.4	-0.4	-0.4	-0.3	-0.3	-0.3	-0.3	-0.2	-0.1		
Other debt creating or reducing flow (please specify)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual		**10.4**	**1.2**	**1.3**	**8.7**	**-5.8**	**2.1**	**2.0**	**2.6**	**5.7**	**1.5**	**-2.5**	**4.5**	**1.3**	**1.8**	**2.1**	**1.1**	**1.0**	**0.0**	**3.0**	**1.3**
Sustainability indicators																					
PV of public debt-to-GDP ratio 3/	...	...	...	...	...	...	...		...	...	**48.2**	**44.6**	**45.0**	**42.5**	**40.9**	**39.4**	**36.7**	**28.2**	**19.1**		
PV of public debt-to-revenue and grants ratio	...	...	...	...	...	...	...		...	...	**227.9**	**280.1**	**244.6**	**235.9**	**231.5**	**219.9**	**205.8**	**159.6**	**125.1**		
Debt service-to-revenue and grants ratio 4/		**3.6**	**5.0**	**7.1**	**29.1**	**31.6**	**33.8**	**27.1**	**34.1**	**20.3**	**15.0**	**69.0**	**62.4**	**73.0**	**72.1**	**74.9**	**89.7**	**69.4**	**32.0**		
Gross financing need 5/	4.0	-2.5	-0.4	2.2	7.8	5.8	2.3	1.2	1.1	1.1	0.1	9.3	13.6	12.8	13.1	13.7	15.0	11.4	5.5		
Key macroeconomic and fiscal assumptions																					
Real GDP growth (in percent)	13.6	0.1	8.8	5.8	6.9	1.8	-5.6	-2.4	2.3	3.0	-1.6	0.6	2.2	3.1	3.6	3.8	3.7	2.6	2.9	**1.9**	**3.1**
Average nominal interest rate on external debt (in percent)		1.8	3.5	3.3	3.6	3.3	6.3	4.0	2.7	2.7	2.2	1.8	1.8	1.5	1.4	1.3	1.1	0.9	0.9	**3.3**	**1.5**
Average real interest rate on domestic debt (in percent)	0	-6.2	-1.3	6.2	2.2	10.6	2.1	2.3	-0.5	-0.1	3.6	-5.2	1.2	1.0	1.1	1.3	1.4	2.3	3.8	**1.9**	**1.0**
Real exchange rate depreciation (in percent, + indicates depreciation)		-5.5	1.0	1.7	12.3	24.4	5.4	-8.5	4.1	2.7	-5.6	...	...	...	...	...	...	...	...	**3.2**	**...**
Inflation rate (GDP deflator, in percent)	6.2	8.4	1.3	-4.2	0.8	-8.0	-1.1	-0.8	2.3	1.5	-2.1	6.3	1.4	1.8	2.1	2.4	2.5	3.1	2.3	**-0.2**	**2.8**
Growth of real primary spending (deflated by GDP deflator, in percent)		-8.5	17.2	0.5	2.7	-20.8	-29.9	5.0	-5.7	12.8	32.6	-2.9	-2.3	4.2	2.4	1.6	1.8	3.6	-2.2	**0.6**	**2.4**
Primary deficit that stabilizes the debt-to-GDP ratio 6/		...	...	...	...	...	...	...	-1.8	-4.7	-1.8	4.9	-3.1	0.5	0.2	-0.1	0.8	0.1	0.4	**-2.8**	**0.3**
PV of contingent liabilities (not included in public sector debt)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central, state, and local governments, central bank, government-guaranteed debt. Definition of external debt is Currency-based.

2/ The primary balance assumes debt relief under the CCRT as a capital grant (subject to availability of resources).

3/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

5/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

6/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

7/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 3. Chad: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2021–2031

(In percent)

	Projections 1/										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of debt-to GDP ratio											
Baseline	21.6	22.2	22.2	21.5	19.9	18.7	18.2	17.4	17.0	16.9	16.9
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	21.6	23.9	25.0	25.3	25.6	26.1	27.7	28.8	**30.5**	**32.6**	**34.7**
B. Bound Tests											
B1. Real GDP growth	21.6	24.0	26.1	25.4	23.5	22.0	21.5	20.5	20.1	20.0	19.9
B2. Primary balance	21.6	22.1	22.7	21.7	19.1	17.1	16.2	14.8	14.3	14.2	14.0
B3. Exports	21.6	28.5	**38.2**	**37.2**	**35.1**	**33.3**	**32.6**	**31.4**	**30.4**	29.2	28.0
B4. Other flows 3/	21.6	25.1	27.3	26.6	24.8	23.5	22.9	22.0	21.3	20.8	20.3
B6. One-time 30 percent nominal depreciation	21.6	28.0	23.9	23.3	21.4	19.9	19.4	18.4	18.0	18.2	18.4
B6. Combination of B1-B5	21.6	29.6	**31.2**	**30.3**	28.3	26.7	26.1	25.0	24.1	23.6	23.1
C. Tailored Tests											
C1. Combined contingent liabilities	21.6	20.7	19.6	18.1	15.4	13.4	12.5	11.3	10.8	10.8	10.8
C2. Natural disaster	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
C3. Commodity price	21.6	26.3	30.0	29.3	27.3	25.5	24.6	23.1	21.9	20.8	19.8
C4. Market Financing	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
Threshold	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**	**30**
PV of debt-to-exports ratio											
Baseline	65.6	70.1	73.0	73.5	68.4	65.6	64.9	62.8	62.4	63.9	66.7
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	65.6	75.2	82.4	86.4	87.9	91.7	98.7	104.3	111.9	123.1	137.1
B. Bound Tests											
B1. Real GDP growth	65.6	70.1	73.0	73.5	68.4	65.6	64.9	62.8	62.4	63.9	66.7
B2. Primary balance	65.6	69.6	74.9	74.0	65.5	60.0	57.6	53.6	52.4	53.4	55.4
B3. Exports	65.6	110.4	**195.3**	**196.7**	**186.3**	**181.3**	**180.0**	**176.1**	**173.0**	**170.5**	**171.6**
B4. Other flows 3/	65.6	79.2	90.0	90.6	85.1	82.3	81.6	79.5	78.2	78.4	80.3
B6. One-time 30 percent nominal depreciation	65.6	70.1	62.7	63.1	58.2	55.4	54.8	52.8	52.4	54.6	57.9
B6. Combination of B1-B5	65.6	98.0	89.1	118.9	111.6	107.8	106.9	104.1	101.7	102.1	104.6
C. Tailored Tests											
C1. Combined contingent liabilities	65.6	65.1	64.7	61.7	52.7	47.0	44.6	40.8	39.7	40.8	42.5
C2. Natural disaster	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
C3. Commodity price	65.6	99.2	115.5	113.5	102.9	95.5	90.4	86.2	82.9	80.9	80.9
C4. Market Financing	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
Threshold	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**	**140**
Debt service-to-exports ratio											
Baseline	6.8	7.7	7.6	8.2	8.7	9.1	6.7	7.9	6.0	5.1	5.4
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	6.8	8.0	8.5	9.7	**10.9**	**12.2**	9.6	**11.9**	10.0	9.0	9.7
B. Bound Tests											
B1. Real GDP growth	6.8	7.7	7.6	8.2	8.7	9.1	6.7	7.9	6.0	5.1	5.4
B2. Primary balance	6.8	7.7	7.7	8.3	8.7	9.1	6.7	8.1	6.2	5.5	5.6
B3. Exports	6.8	**10.1**	**13.5**	**15.1**	**15.8**	**16.6**	**12.4**	**14.3**	**13.2**	**15.3**	**15.7**
B4. Other flows 3/	6.8	7.7	7.8	8.4	8.9	9.3	6.9	8.1	7.0	6.7	6.9
B6. One-time 30 percent nominal depreciation	6.8	7.7	7.6	8.1	8.5	9.0	6.6	7.8	5.9	4.1	4.4
B6. Combination of B1-B5	6.8	8.8	**10.4**	**11.2**	**11.7**	**12.3**	9.1	**10.7**	9.9	8.7	9.1
C. Tailored Tests											
C1. Combined contingent liabilities	6.8	7.7	7.6	8.2	8.6	9.0	6.5	7.7	5.8	4.9	5.2
C2. Natural disaster	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
C3. Commodity price	6.8	9.4	9.5	**10.2**	**10.5**	**10.8**	7.8	9.0	8.0	7.9	8.0
C4. Market Financing	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
Threshold	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**	**10**
Debt service-to-revenue ratio											
Baseline	**16.9**	**15.7**	**15.4**	**16.4**	**16.8**	**17.4**	13.0	**14.6**	10.8	9.1	9.1
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	**16.9**	**16.3**	**17.0**	**19.3**	**21.1**	**23.3**	**18.4**	**22.0**	**17.9**	**16.0**	**16.4**
B. Bound Tests											
B1. Real GDP growth	**16.9**	**16.9**	**18.1**	**19.4**	**19.8**	**20.5**	**15.3**	**17.2**	12.7	10.7	10.7
B2. Primary balance	**16.9**	**15.8**	**15.5**	**16.6**	**16.9**	**17.4**	12.9	**15.0**	11.1	9.8	9.5
B3. Exports	**16.9**	**16.7**	**17.6**	**19.4**	**19.8**	**20.3**	**15.4**	**17.1**	**15.3**	**17.6**	**17.2**
B4. Other flows 3/	**16.9**	**15.7**	**15.6**	**16.9**	**17.2**	**17.7**	13.3	**14.9**	12.5	11.9	11.7
B6. One-time 30 percent nominal depreciation	**16.9**	**19.7**	**19.4**	**20.3**	**20.9**	**21.5**	**16.0**	**18.1**	13.3	9.2	9.4
B6. Combination of B1-B5	**16.9**	**17.1**	**18.2**	**19.4**	**19.8**	**20.4**	**15.3**	**17.2**	**15.5**	13.5	13.3
C. Tailored Tests											
C1. Combined contingent liabilities	**16.9**	**15.8**	**15.3**	**16.3**	**16.6**	**17.1**	12.6	**14.2**	10.4	8.7	8.7
C2. Natural disaster	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
C3. Commodity price	**16.9**	**19.3**	**19.8**	**21.9**	**21.4**	**21.1**	**15.2**	**16.2**	13.9	13.7	13.2
C4. Market Financing	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
Threshold	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**	**14**

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Chad: Sensitivity Analysis for Key Indicators of Public Debt, 2021–2031

(In percent)

	Projections 1/										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of Debt-to-GDP Ratio											
Baseline	**44.6**	**45.0**	**42.5**	**40.9**	**39.4**	**36.7**	34.6	32.5	30.7	29.2	28.2
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	**45**	**48**	**47**	**48**	**50**	**50**	**51**	**51**	**52**	**52**	**53**
B. Bound Tests											
B1. Real GDP growth	**45**	**231**	**249**	**246**	**242**	**237**	**233**	**230**	**226**	**223**	**222**
B2. Primary balance	**45**	**47**	**47**	**45**	**44**	**42**	**39**	**37**	**35**	33	32
B3. Exports	**45**	**49**	**55**	**53**	**51**	**48**	**46**	**43**	**41**	**38**	**36**
B4. Other flows 3/	**45**	**48**	**48**	**46**	**44**	**41**	**39**	**37**	35	33	32
B6. One-time 30 percent nominal depreciation	**45**	**274**	**268**	**261**	**254**	**246**	**240**	**234**	**228**	**223**	**219**
B6. Combination of B1-B5	**45**	**45**	**47**	**45**	**45**	**44**	**42**	**40**	**39**	**37**	**35**
C. Tailored Tests											
C1. Combined contingent liabilities	**45**	**59**	**56**	**54**	**53**	**49**	**47**	**44**	**41**	**39**	**37**
C2. Natural disaster	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
C3. Commodity price	**45**	**220**	**223**	**227**	**229**	**227**	**225**	**222**	**218**	**216**	**214**
C4. Market Financing	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
Public debt benchmark	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**	**35**
PV of Debt-to-Revenue Ratio											
Baseline	280.1	244.6	235.9	231.5	219.9	205.8	199.1	184.3	171.3	165.8	159.6
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	280	258	259	268	273	274	281	278	275	277	277
B. Bound Tests											
B1. Real GDP growth	280	1240	1342	1349	1311	1292	1305	1264	1228	1234	1224
B2. Primary balance	280	256	261	257	247	233	227	211	195	188	179
B3. Exports	280	269	303	298	284	269	263	245	229	218	206
B4. Other flows 3/	280	260	264	260	247	233	226	210	195	188	179
B6. One-time 30 percent nominal depreciation	280	1511	1503	1496	1436	1400	1400	1340	1287	1282	1259
B6. Combination of B1-B5	280	245	261	254	248	244	242	227	215	208	199
C. Tailored Tests											
C1. Combined contingent liabilities	280	320	311	307	293	277	268	248	229	219	209
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	280	1390	1439	1492	1412	1354	1326	1236	1201	1207	1197
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	69.0	62.4	73.0	72.1	74.9	89.7	86.2	83.2	80.2	74.8	69.4
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2039 2/	69	64	76	77	81	104	103	104	104	102	98
B. Bound Tests											
B1. Real GDP growth	69	68	84	85	88	104	102	101	97	93	88
B2. Primary balance	69	63	72	76	73	85	82	81	78	75	70
B3. Exports	69	62	73	73	76	90	87	84	83	81	75
B4. Other flows 3/	69	62	73	73	75	90	86	83	82	77	72
B6. One-time 30 percent nominal depreciation	69	62	73	70	75	89	84	83	77	71	65
B6. Combination of B1-B5	69	62	74	75	76	91	87	84	81	76	71
C. Tailored Tests											
C1. Combined contingent liabilities	69	63	65	67	70	83	81	80	77	73	69
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	69	75	88	89	98	112	106	99	95	91	87
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.